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PROJECT REPORT

BRIDGEFINITY: Turning Intentions Into Impact

A Project Report submitted in partial fulfillment of the requirements for the degree of

BACHELOR OF TECHNOLOGY
in
COMPUTER SCIENCE AND ENGINEERING
(III Semester)

Submitted By

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Group No – G2

Guided by
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Department of Computer Science and Engineering

Jhulelal Institute of Technology, Nagpur

Session: 2026–27

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PROGRAM EDUCATIONAL OBJECTIVES (PEOs)

PEO 1 – To develop an ability to analyze software, understand technical specifications, design and provide novel engineering solutions and efficient product designs.
PEO 2 – To develop professional skills that prepare students for immediate employment and life-long learning in advanced areas of computer science and related fields.
PEO 3 – To develop necessary communication skills to bridge the gap between advanced technology and end users in practice of computer science products.
PEO 4 – To develop technical skills to adapt to an ever-changing professional environment.
PSO 1 – Students will be able to use professional engineering practices for development, operation and maintenance of software.
PSO 2 – Students will have the ability to understand, analyze and develop computer programs related to algorithms, system software, data structures, web design and networking.

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CERTIFICATE

This is to certify that the report entitled "BRIDGEFINITY: Turning Intentions Into Impact" submitted to Rashtrasant Tukadoji Maharaj Nagpur University, Nagpur is the bonafide work of the students of 3rd semester in partial fulfillment of the requirement for the degree of Bachelor of Technology in Computer Science and Engineering. The work was carried out under supervision at the Department of Computer Science and Engineering, Jhulelal Institute of Technology, Nagpur during academic year 2026–27.

Prof. Guide Name – Asst. Prof., CSE, JIT, Nagpur.
Dr. Uma Patel (Thakur) – HOD, CSE, JIT, Nagpur.

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DECLARATION

We hereby declare that the work for the report entitled "BRIDGEFINITY: Turning Intentions Into Manasvi Dupare Impact" has been carried out under supervision at the Department of Computer Science and Engineering, Jhulelal Institute of Technology, Nagpur during academic year 2026–27. The work is original and has not been submitted earlier for the award of any degree at another Institution/University.

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ACKNOWLEDGEMENT

We express our sincere gratitude to our guide, Assistant Prof. Shubhangi Doye, for valuable guidance, encouragement and support throughout the project. We also thank Dr. Uma Patel (Thakur), HOD, CSE, the institute, all teaching and non-teaching staff, and our family members and friends for their support.
Their suggestions and encouragement helped us understand the practical aspects of requirement analysis, system design, documentation and presentation. The project also provided an opportunity to connect computer science concepts with a community-oriented problem.

ABSTRACT

BRIDGEFINITY: Turning Intentions Into Impact is a proposed centralized web platform designed to connect verified Non-Governmental Organizations (NGOs) with individuals interested in volunteering. Information about NGOs, volunteering opportunities and social events is often distributed across social media pages, messaging groups, websites and personal contacts, making it difficult for first-time volunteers to identify genuine organizations and suitable opportunities.

The proposed platform provides a single organized place where users can explore NGO profiles, search opportunities by location and area of interest, view upcoming events, register for volunteering activities and receive relevant updates. NGOs can present their activities, publish events and reach a wider audience of potential volunteers.

The system emphasizes usability, accessibility, organized communication and transparent volunteer management. By bringing important information into one platform, BRIDGEFINITY aims to reduce the effort required to discover opportunities and improve the connection between people willing to contribute and organizations working for community welfare.

The project follows requirement analysis, system design, development, testing, deployment and documentation. The modular approach allows the platform to be extended in the future with features such as skill-based matching, mobile support, automated notifications, analytics and multilingual access.

INTRODUCTION

Volunteering plays a significant role in creating positive social impact by bringing together individuals who are willing to contribute towards community welfare. Numerous NGOs work in education, environmental conservation, healthcare, animal welfare, disaster relief and social development. These organizations depend on public participation, but potential volunteers may not always know where or how to participate.

At present, volunteering information is frequently scattered across social media pages, WhatsApp groups, websites and personal contacts. This makes it difficult, especially for first-time volunteers, to discover organizations, compare opportunities and participate in events. Information may also become outdated when announcements are posted in multiple places.

BRIDGEFINITY proposes a centralized platform that acts as a bridge between NGOs and volunteers. Users can explore NGO profiles, search opportunities based on location and cause, register for volunteering events and stay updated about upcoming initiatives. NGOs can maintain their information and publish opportunities in a structured way.

The platform is designed with two major user perspectives. Volunteers need simple discovery, filtering and registration, while NGOs need tools to present their work, publish events and manage volunteer participation. A centralized database supports consistent storage and retrieval of these records.

1.1 Aim: To develop a centralized platform that connects NGOs with volunteers by providing verified

NGO information, volunteering opportunities, event updates and an easy registration process, making community participation more accessible and organized.

1.2 Objectives: Develop a centralized database of verified NGOs; enable search based on location and area of interest; provide detailed NGO profiles and upcoming volunteering opportunities; simplify volunteer registration; improve communication between NGOs and volunteers; and encourage active participation in community welfare activities.

1.3 Significance: The project demonstrates how a web-based information system can be applied to a community problem. It combines user management, search, event management, database operations and validation into a single practical application.

LITERATURE SURVEY

Existing studies and digital systems show that technology can simplify NGO administration, volunteer discovery and community participation. The reviewed sources provide useful ideas for NGO information management, volunteer matching, web development and location-based services.

Reference 1: Smart NGO Management System, International Journal of Innovative Research in Technology (IJIRT), Vol. 12, Issue 11, April 2026.
Reference 2: NGO Management Platform with Smart Skill-Based Volunteer Matching, International Journal of Advanced Research in Computer and Communication Engineering (IJARCCE), Vol. 14, Issue 12, December 2025.
Reference 3: Mozilla Developer Network (MDN), HTML, CSS and JavaScript Documentation.
Reference 4: Google Maps Platform Documentation.

2.1 Observations: The literature indicates that structured digital records can reduce manual information handling. Search and filtering can improve discovery, while skill-based matching can help connect volunteers with appropriate activities. Web technologies provide a practical foundation for building accessible interfaces.

2.2 Research Gap: There is a need for a simple centralized platform combining verified NGO discovery, opportunity search, event information and volunteer registration in one place. BRIDGEFINITY focuses on bringing these core functions together with a straightforward user experience.

PROBLEM STATEMENT AND OBJECTIVE

The lack of a centralized platform for volunteering makes it difficult for individuals, especially first-time volunteers, to discover genuine NGOs and participate in social initiatives. Information is scattered across social media, WhatsApp groups, websites and personal contacts. Many NGOs also face challenges in reaching new volunteers and promoting events beyond their existing network.
The project therefore focuses on creating a structured digital bridge between volunteers and NGOs. Instead of requiring users to search multiple channels, the platform provides organized profiles, opportunities and event information in one place.
Major functional objectives include volunteer registration and profile management; NGO registration, profile and verification; search by location, cause and opportunity; event publishing and event registration; organized information management; and a simple, responsive and maintainable interface.
Expected benefits include reduced search effort, improved visibility of community initiatives, simpler event registration and better communication between participating users and organizations.

METHODOLOGY

The project follows a phased methodology based on the synopsis. The major phases are Requirement Analysis, System Design, Development, Testing, and Deployment & Documentation.
Phase 1 – Requirement Analysis: study existing volunteering processes, identify problems and collect requirements. Duration: 1 week.
Phase 2 – System Design: define modules, database structure, user flow and interface. Duration: 1 week.
Phase 3 – Development: develop frontend, backend/database and major platform features. Duration: 3 weeks.
Phase 4 – Testing: test modules, fix bugs and verify usability. Duration: 1 week.
Phase 5 – Deployment & Documentation: deploy, prepare documentation/user manual and demonstrate. Duration: 1 week.
During requirement analysis, the team studies current volunteering processes and identifies the information needed by both volunteers and NGOs. The design stage converts these requirements into modules, data entities and user flows. Development then implements the planned functionality, followed by testing and documentation.
Project activities include conducting literature review, studying current volunteering processes, interacting with volunteers and community organizations, collecting and analyzing requirements, comparing existing solutions and identifying limitations, and designing, developing and testing the proposed platform.

DATA FLOW DIAGRAM

The DFD represents the major information flow in BRIDGEFINITY. Volunteers and NGOs interact with the platform, while the application handles searching, registration, event information and management. The database stores user, NGO, event and registration records.

Volunteer → Platform: registration details, search criteria and event registration.

NGO → Platform: organization profile, verification information and event details.

Platform → Volunteer: NGO profiles, search results, event information and registration status.

Platform → NGO: volunteer registrations and event participation information.

Platform ↔ Database: storage and retrieval of user, NGO, event and registration information.

The data flow can be understood as a cycle. A user provides input through the interface, the application validates and processes that input, relevant information is retrieved or stored in the database, and the result is presented back to the appropriate user. This separation helps keep the system organized and maintainable.

IMPLEMENTATION

BRIDGEFINITY is planned as a modular web application. The frontend provides the user interface, the application layer processes requests and business rules, and the database stores structured information.

Frontend: HTML is used for structure, CSS for layout and presentation, and JavaScript for client-side interaction. These technologies provide the basic foundation for responsive pages and interactive controls.

NGO Module: organizations can maintain profiles and publish volunteering opportunities including location, cause, description, contact details and events. The profile gives volunteers a structured view of the organization's work.

Volunteer Module: users can create profiles, discover NGOs and opportunities, filter information by location or cause, view event details and register. This module focuses on reducing the effort required to find suitable activities.

Event Management: NGOs can publish event title, date, location, cause, description and registration information. Event records can then be displayed to volunteers for discovery and registration.

Database and Security: records are maintained for volunteers, NGOs, events and registrations. Input validation and role-based access should be used. Validation reduces incorrect or incomplete data, while role-based access helps ensure that users access only the functions appropriate to their role.

6.1 User Flow: A volunteer creates or accesses a profile, searches for an NGO or opportunity, applies filters if required, views event details and registers. An NGO maintains its profile, publishes an event and reviews volunteer registration information.

6.2 Maintainability: A modular structure makes it easier to test individual components, fix errors and add future features without redesigning the complete system.

HARDWARE & SOFTWARE REQUIREMENT

Hardware: Intel Core i3 or equivalent processor, minimum 4 GB RAM, at least 10 GB free storage, standard laptop/monitor and internet connection.
Software: Windows/Linux operating system, HTML, CSS, JavaScript, selected backend technology, selected database, Visual Studio Code or equivalent editor, Chrome/Firefox/Edge browser and MS Word or equivalent for documentation.
Optional location services: Google Maps Platform can be integrated if location-based map functionality is enabled.
7.1 Development Environment: The system can be developed using a standard student-level computer with a modern web browser and code editor. Internet connectivity is useful for development, testing, documentation and optional map-based services.
7.2 Requirement Rationale: The listed requirements are intended to keep the proposed system accessible for development and demonstration while leaving scope for future scaling.

TESTING

Testing is required to verify that the proposed system behaves correctly, stores information consistently and provides an understandable user experience. Testing is performed against important user actions and expected results.
Test 1 – Volunteer registration with valid details: profile should be created.
Test 2 – Missing required registration field: validation message should appear.
Test 3 – NGO search by location/cause: relevant records should be displayed.
Test 4 – Event listing: selected event details should display correctly.
Test 5 – Event registration: registration should be recorded.
Test 6 – NGO event creation: new event should be stored and listed.
Test 7 – Invalid input: invalid data should be rejected.
Test 8 – Navigation: major pages should open correctly.
8.1 Testing Approach: Functional testing checks whether each feature performs its intended task. Validation testing checks incomplete and invalid inputs. Navigation testing checks whether users can move between major pages without errors. Database-related testing checks whether important records are stored and retrieved consistently.
8.2 Usability Consideration: The interface should use clear labels, understandable messages and simple navigation so that first-time users can discover opportunities without extensive instructions.

RESULTS / COMMUNITY ENGINEERING PROJECT OUTCOME

The proposed BRIDGEFINITY platform is designed to create a more organized connection between NGOs and volunteers. The expected outcomes are directly aligned with the project aim and synopsis. Develop a centralized platform connecting NGOs with volunteers efficiently. Make volunteering opportunities more accessible to the public. Increase the visibility of NGOs and their social initiatives. Simplify discovering and registering for volunteer activities. Improve communication between NGOs and volunteers.

Encourage community participation and social responsibility. Promote organized and transparent volunteer management. Support education, environment, healthcare and animal welfare causes.

9.1 Community Impact: By reducing information fragmentation, the platform can make it easier for interested individuals to discover activities. NGOs can use structured event information to reach potential volunteers beyond their existing contacts.

9.2 Learning Outcome: The project provides practical exposure to requirement analysis, web interface development, data organization, validation, testing, documentation and presentation. It also demonstrates how software engineering can be applied to a community-oriented objective.

CONCLUSION AND FUTURE SCOPE

BRIDGEFINITY addresses the practical problem of connecting volunteers with genuine NGOs and suitable social activities when information is distributed across multiple channels. The centralized platform brings NGO information, volunteering opportunities, event updates and registration into one organized system.

The project combines software engineering practices with a community-oriented objective. By simplifying discovery and communication, the system can make volunteering more accessible and help NGOs reach people willing to contribute.

The proposed system provides a foundation that can be enhanced as requirements grow. Future development can focus on skill-based volunteer matching, mobile application support, automated notifications, advanced NGO verification and trust indicators, volunteer participation history and certificates, analytics dashboards, location-based maps and route assistance, multilingual interface, and integration with more community organizations.

Skill-based matching could compare volunteer interests or skills with opportunity requirements. Mobile support could make event discovery more convenient. Notifications could remind users about registrations and upcoming events. Analytics could help NGOs understand participation patterns while supporting better planning.

Overall, BRIDGEFINITY demonstrates a practical approach to organizing volunteer information and strengthening the connection between community organizations and individuals who want to contribute.

PO / PSO ATTAINMENT

Indicative attainment scale: HIGH = 3, MEDIUM = 2, LOW = 1.
PO1: 3 | PO2: 3 | PO3: 3 | PO4: 2 | PO5: 3 | PO6: 3 | PO7: 2 | PO8: 3 | PO9: 3 | PO10: 3 | PO11: 2 | PSO1: 3 | PSO2: 3.
Final attainment may be revised by the department according to its evaluation methodology.
The project supports the attainment of program outcomes through problem identification, software design, implementation, testing, documentation, communication and teamwork. The exact departmental evaluation remains the final basis for attainment.

REFERENCES

[1] Smart NGO Management System, International Journal of Innovative Research in Technology(IJIRT), Vol. 12, Issue 11, April 2026.

[2] NGO Management Platform with Smart Skill-Based Volunteer Matching, International Journal of Advanced Research in Computer and Communication Engineering (IJARCCE), Vol. 14, Issue 12, December 2025.

[3] Mozilla Developer Network (MDN), HTML, CSS and JavaScript Documentation.

[4] Google Maps Platform Documentation.

[5] Primary field interaction conducted with Vrikshit Foundation and Orange City Volunteers duringproject requirement analysis (5th June–21st June 2026).